

STRUCT MEMORY-R1: Overview

Multi-Session Dialogue

A: I visited Tokyo last summer
B: Did you see Shibuya?
A: Yes, and Mt. Fuji too

d_t

1



Memory Manager

RL-TRAINED

Updates tree memory from new dialogue turns

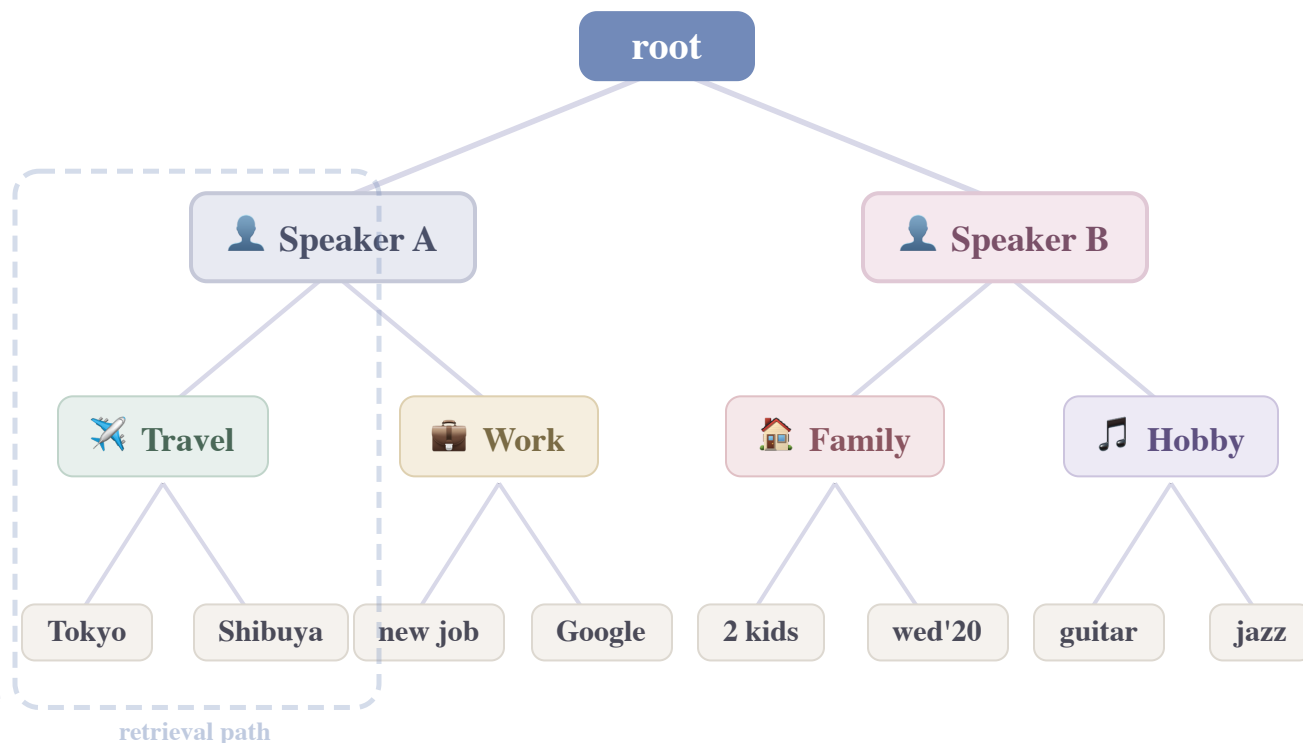
ADD UPDATE DELETE
NOOP CREATE_TOPIC
CREATE_NODE MOVE
SPLIT MERGE

write



Tree-Structured Memory

root → speaker → topic → fact



Example Retrieval Plan: "Where did A travel?"

SELECT_SPEAKER: A → SELECT_TOPIC: Travel → SELECT_IDS: all →
STOP → [Tokyo, Shibuya, Mt. Fuji]



Retrieval Agent

RL-TRAINED

Navigates tree to find relevant memories

SELECT_SPEAKER SELECT_TOPIC
SELECT_ENTITY SELECT_IDS STOP

?

Question

"Where did A travel?"

evidence



Answer Agent

RL-TRAINED

Reasons over retrieved memories to produce answer

SELECT_MEMORY REASON
GENERATE_ANS



Retrieved Mem

[m3] Shibuya
[m7] Mt. Fuji
[m12] Shinjuku

answer



Final Answer "Tokyo, Shibuya, Mt. Fuji"